한양대학교
HANYANG UNIVERSITY

Graduation Project Proposal

Title	OMRiDA: Omni Mathematical expression Recognition via Domain Adaptation
Summary (in 150 words)	Mathematical expression recognition (MER) is a practical and meaningful task, enabling users to handle mathematical expressions (ME) as easily as they do with regular text - such as copying text from an image for search or writing in a document. However, compared to standard text recognition, MER is more challenging due to the complex two-dimensional layouts of mathematical symbols. Previous approaches typically trained models on either printed or handwritten ME datasets, but such models often fail to generalize well across domains. In this research, we aim to build a generalized MER model that performs reliably on both printed and handwritten ME data. By applying domain adaptation techniques, we attempt to bridge the domain gap and enhance generalization. Our goal is to develop a robust model that works well on both printed and handwritten ME datasets and achieves state-of-the-art performance, particularly on the offline handwritten MER dataset.
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Contents

1. Project background and goals

From old times, humans left their records on the document. With the development of big data and machine learning technology, we built models that can automatically understand what we wrote and change it into digital form. Recently, optical character recognition (OCR) and scene text recognition (STR) technology have developed so far that we can extract text directly from images on our smartphones. For example, on iPhones, we can simply press the text part of the image to activate the automatic recognition system to extract in the form of text. However, this STR technology on most smartphones is limited to text and they can't recognize mathematical expressions (MEs). Our project idea started from this point that we need mathematical expression recognition (MER) technology on our smartphones since we also use mathematical documents a lot, especially for students.

According to **[1],** MER is different from OCR tasks by the presence of a two-dimensional (2D) structure and the number of symbols, which is more than 1500. Also, these symbols are often very similar to each other or hard to recognize, especially for handwritten mathematical expressions (HME). Furthermore, there're various HME challenges such as writing multiple symbols side by side, peculiarities of each person's handwriting, the spatial relations classification, and an implicit multiplication operator between subexpressions. It's true that many handwritten mathematical expressions recognition (HMER) technologies have been studied and some of them are already commercialized as online services. Despite these new developments in HMER, much more research is still needed to enhance their performance, make them recognize longer and complex MEs, and reduce the errors. As shown in many researches, the highest state-of-the-art (SOTA) accuracy of standard STR tasks is higher than 90% but the accuracy of HMER tasks is only around 70%. In case of benchmark datasets, the SOTA accuracy of STR on ICDAR2015 is 93.5% from DTrOCR 105M **[2].** On the other hand, the SOTA accuracy of HMER on CROHME 2014 is 61.23% from TAMER **[3]** and one on HME100K is 69.51% from PosFormer **[4]**.

There're various data types among MEs such as offline or online and handwritten or printed. Offline is written on real-life backgrounds such as paper, notes, boards, and so on. Online is written on screen using fingers or digital pens. They include stroke information since they can store the whole process of writing. Handwritten is written with a human hand, both offline and online. Printed is digitally made using LaTex, which is a software system for typesetting documents. Most MEs are converted into LaTex format to be expressed in specific format and to be labeled. According to each data type, the level of recognition differs. Among them, offline HMER is a challenging task since they need different approaches from printed or online HMER. We chose to study both offline HMER and printed mathematical expression recognition (PMER) because the model which can recognize all kinds of MEs at once will be useful considering real-life applications. For instance, when we want to know whether the process of solving mathematics problems is right, we can extract printed questions and handwritten MEs at once using smartphones just like we've done with text.

Previously, PMER and HMER models worked separately since they're developed and trained on different datasets. What it means is that most PMER models are trained on printed mathematics expression (PME) dataset and most HMER models are trained on HME dataset. It's because PMER and HMER have similar approaches for recognition in a large framework but the features of the dataset aren't similar, requiring different techniques in detail. Also in CROHME competition, they let HMER models use only CROHME dataset, which is composed of only HME data **[5]**. However, the model we need in real-life is one that can recognize both PME and HME well, even if they're mixed in one image. Getting ideas from the domain gap between PMER and HMER, we're planning to apply domain adaptation (DA) from PMER to offline HMER. Using common features and relationships between them, we'll build a model that is robust in both PMER and HMER and overcome the domain gap. Finally, we'll try to exceed SOTA for HMER tasks with a domain adaptation approach.

2. Project content

This project aims to achieve decent performance in extracting LaTex representations of mathematical expressions from input images containing both handwritten and printed mathematical expressions. MER is a complex sub-task of STR, due to the inherent complexity of mathematical expressions, which contain two-dimensional spatial structures such as fractions, superscripts and subscripts, matrices, and integrals, unlike typical one-dimensional text sequences. To address such special difficulty, leveraging the relationship between printed and handwritten mathematical expressions by applying domain adaptation techniques on existing models trained from printed ME dataset was proposed. To realize this objective, the detailed contents of the project are outlined as follows:

2.1. Understanding general concepts of related topics

- Explore benchmark datasets, evaluation metrics, and model pipelines for STR, PMER, and HMER tasks.
- Establish a clear understanding of the similarities and distinctions among STR, PMER, and HMER to guide model selection and adaptation strategies.

2.2. Survey of existing Methodologies

- Review both traditional and SOTA approaches to STR, PMER, and HMER tasks.
- Understand how these models have evolved and how the approaches are different, focusing on which methodologies have been employed to handle special challenges in ME domain.

2.3. Baseline model Selection and Implementation

- Select baseline SOTA models for both printed and handwritten MER task.
- Implement and evaluate these models on benchmark datasets to serve as reference baselines. The details of the implementation plan with the objectives at this stage are as follows:

1) Evaluate the performance of existing HMER models on HME datasets. These results will serve as a criteria to check how our model works well compared with the previous existing models.
2) Check the domain gap between handwritten and printed domain. Our hypothesis is that PMER model would work poorly on HME datasets, and HMER model would not work well on PME datasets. Checking the disparity of the two domains, we primarily

aim to bridge this gap through DA techniques and build a model that both works well on both PME dataset and HME dataset, and secondly will try to see if the mitigation of the domain gap would lead the SOTA performance on HME dataset. In this perspective, the baseline implementation on this stage would be used as a reference to see how much our model bridges the existing domain gap.

2.4. DA

- Study domain adaptation techniques with the goal of shifting representative features from printed ME domain to handwritten ME domain

2.5. Performance Evaluation and Comparison

- Assess the performance of the proposed model in this project with evaluation metrics for HMER task and PMER task respectively. Our ideal goal is to show that our model performs well on both tasks unlike previous models, which only show decent performance on a single task. Considering that PME and HME have different characteristics but they share the common characteristic as ME, we'll ultimately try to reach better performances on both datasets compared with the baseline model results from each domain. Ideally, the performance of our model on HME datasets should be better than that of the PMER model, and the performance of our model on PME datasets should be better than that of HMER model. Furthermore, we'll consider the possibility that our model may even perform better on HME dataset than the existing SOTA HME model which is built and trained only for the HMER task.
- Compare the result of our approach with other prior researches that utilized both two domains. The main research keyword in this project is to utilize PME domain and HME domain together, and there have been similar approaches such as **[6]**, **[7]**, **[8]**. We will first see how the performance of our model differs from these approaches, and then analyze why and how each approach made the different results.

2.6. Improving model

- Conduct additional experiments to improve the model. In this process, we will both try to reduce the domain gap and try to improve the model performance to reach SOTA performance, by mainly considering hyperparameter tuning, ablation studies, trying alternative DA methods, model ensemble, and data augmentation.
- In particular, we are considering adding realistic background templates (e.g., blackboard, grid, crumpled paper, newspaper, etc.) on existing handwritten dataset to make robust model by training on real-world environments.

3. Plan

Monthly goals and content were organized under the whole project contents written on section 2. The tentative monthly goal and progress schedule along with division of roles are as follows:

3.1. March - Project objective confirmation & Understanding general concepts of related topics

1) Project Topic Selection

- To come up with various possible project topics and get consult from advisor professor. Sign language, OCR, question generation topics were proposed and OCR was chosen as a project domain, considering the environment limitations (gpu, deadline, etc.)
- To study OCR related topics and concepts. Starting from the OCR survey paper, several papers regarding scene text detection (STD), STR, MER tasks were reviewed. Further studies including benchmark dataset, evaluation metric, several models and approaches on each task were proceeded to fully understand text recognition domain.

2) Project Objective Confirmation

- After understanding the outline of text recognition topics, we've found out that current text recognition models achieve near-complete proficiency on single language text domain while it loses its performance on multilingual text images and HME images. Considering the discovered challenges, 4 possible specific objectives were proposed for text recognition topic:

① Apply domain adaptation technique from PME domain to HME domain.

② Augment existing benchmark dataset by changing the background of original ME images into realistic background templates (e.g., blackboard, grid, crumpled paper, newspaper, etc.) to make robust HMER model.

③ Develop text-type agnostic model which can extract textual contents from an image, even when the image contains lots of various text contents (e.g. multilingual texts, numbers, special symbols).

④ Improve scene text editing (STE) model that can modify the desired part of the text in the image while keeping the natural accordance between modified parts and the original parts.

- Among them, improving HMER model with DA techniques for PME domain model was decided as the specific project objective after consultation with the advisor professor.

3) Division of Roles
- Both members studied the field of text recognition task equally and shared what he/she learned.
- Each student in charged of studying different papers and shared the contents of each paper

3.2. April - Survey of existing methodologies & Baseline model selection and implementation (1)

1) Select baseline SOTA models for both printed and handwritten MER task.
2) Implement and evaluate models on benchmark datasets to establish performance baselines.
3) Submit project proposal.
4) Division of Roles
- Each student is assigned to reproduce PMER baseline model and HMER baseline model, respectively.

3.3. May - Baseline model selection and implementation (2) & DA

1) Keep implementing baseline models if it was not finished in April.
2) Study DA techniques and choose possible DA methodologies for our project (shifting representative features from PME domain to HME domain to reach SOTA performance).
3) Apply selected DA techniques to our baseline model.
4) Division of Roles
- Each student is assigned to apply distinct DA technique to the same baseline model

3.4. June - Performance evaluation and comparison

1) Assess the performance of our model in this project with evaluation metrics for HMER task on various HMER benchmark datasets.
2) Compare the result with baseline SOTA model and other previous approaches using both printed and handwritten domains (Similar approach with ours).
3) Submit Project Progress Report.
4) Division of Roles
- Dividing HMER benchmark datasets, and each student is assigned to apply our model on different benchmark datasets.

3.5. July - Review

1) Review past progress and achievements. Check if any mistakes or missing were made, and fix if needed.
2) Modify future direction and schedules considering current progress.
3) Division of Roles
- To be determined. All the members will be engaged in the project equally.

3.6. August - Improving model

1) Conduct additional experiments to improve model the model, including:
- Hyperparameter tuning
- Ablation studies
- Trying alternative DA methods
- Model ensemble
- Data augmentation
2) If time allows, establish our own HMER dataset for the robust model, by adding realistic background templates (e.g., blackboard, grid, crumpled paper, newspaper, etc.) on existing HMER datasets.
3) Division of Roles
- To be determined. All the members will be engaged in the project equally.

3.7. September - Dissertation paper preparation

1) Aggregate project progress and results. Summarize key findings and contributions of our project.
2) Compare the final project result with the initial objective, and analyze the reason why they became different, if needed.
3) Suggest future improvement directions for our project.
4) Develop the dissertation paper.
5) Division of Roles
- To be determined. All the members will be engaged in the project equally.

3.8. October - Final Presentation preparation

1) Prepare final presentation materials such as PPT, poster, demo video, presentation script, etc.
2) Upload research materials on github.
3) Division of Roles
- To be determined. All the members will be engaged in the project equally.

4. Conclusion

4.1. Technical impacts and possible future extensions

Through applying domain adaptation from PMER to HMER, we'll try to build a model that can recognize both PME and HME, and achieve SOTA performance with our proposed model on HMER tasks. If our goal is achieved, we can recognize general ME (both PME and HME) using only one model instead of applying separate PMER model and HMER model to each task. This simplification of the process - reducing the requirement from two separate models to a single unified model will allow us to develop lightweight mobile services that can be implemented as on-device functions. With such services, the image recognition function can be extended, enabling users to extract all types of MEs from an image simply by pressing on the ME part, just as they do when extracting text or generating stickers (e.g., removing backgrounds using instance segmentation) on mobile devices.

Another advantage of reducing the domain gap between PMER and HMER is that we can use a massive amount of PME dataset to train HMER model. It's a time-efficient and cost-efficient way because collecting PME dataset is relatively easier than HMER dataset. We can find PME on LaTex documents or digital textbooks while we have to collect HME manually. Furthermore, HME has various handwriting and structure according to the writer, making it difficult and expensive to give accurate labels to them. So, it's efficient to use both the large amount of PME dataset and HME dataset to train MER models **[9]**.

From an academic point of view, our approach to bridge the gap between the handwritten and printed domains in the ME field can be extended to similar fields such as graphs, charts, tables, musical scores, electrical schematics, chemical diagrams, and so on. With further studies in these fields, users could easily generate academic visual content on a computer without requiring professional skills, simply by sketching what they intend to create on a blank sheet of paper and then using the image of the sketch as an input to the model.

4.2. How it's related to future career path - Sungmin Yang

My research interests lie primarily in multimodal AI and generative AI within the field of Computer Vision. I'm particularly interested in leveraging visual AI techniques to build real-world applications and develop useful services. In this context, the HMER task stands out as a highly promising area with clear practical value.

I expect to dive into the real-world applications of multimodal AI and elucidate deep learning concepts while proceeding the project. From the previous undergraduate courses and personal experiences, I have implemented several machine learning architectures, but they were mainly for classic tasks and basic implementations rather than being closely aligned with the real-world applications. Through this project, I personally aim to advance my deep learning knowledge and coding skills by conducting research dealing with the real world demand. The experience of devising own model for practical usage and combining, changing, utilizing previous models with several steps of experimenting and improving the model to achieve specific goal and summarizing the full scope of the research into an organized paper and presentation materials will be an valuable asset and solid groundwork before embarking on the research experience as a graduate student in my research interest field.

4.3. How it's related to future career path - Soyeon Kim

I'm interested in the computer vision domain, so I've studied this domain, especially object detection and segmentation. I've also done laboratory practice of intelligence computing, studying anomaly object detection and few-shot semantic segmentation. As a senior this year, I'm aiming to proceed with a project within the field of object detection or segmentation. In detail, my goal is to carry out projects on topics that can be made into real-life services and to improve practical skills through this development experience. Among the wide computer vision domain, STR includes both object detection and segmentation with various approaches. So, I think this project will be helpful to experience developing interesting areas, gain professional knowledge, and develop skills in the computer vision domain.

If possible, I also want to study more about STE, which is a recently started research area and includes both STR and generative AI. Since the recent trend is generative AI and many corporations are trying to provide generative AI services, it would be helpful to figure out which AI technology is needed for business and people. Domain adaptation is a new area for me to deal with, so it'll help to think about the approach directions of the AI model more diversely. Since I'm planning to get a job after graduation, this one year project on HMER will make me grow up as a developer who can make useful AI techniques.

5. References

In this reference section, we leave some papers of the milestone research for the STR and MER tasks as well as other research works related to our project, by categorizing them into the similar groups. In addition to the papers mentioned in the sections above, we also list some papers that were not previously mentioned, for future reference while proceeding the research. The papers cited in the earlier sections are marked with bold numbers in square brackets. All paper citations are in APA (7th) style. For data only references, we left url links for them. Models are arranged in ascending order of time, alphabetically organized within the same year.

5.1. Survey paper

1) Subramani, N., Matton, A., Greaves, M., & Lam, A. (2020). A survey of deep learning approaches for ocr and document understanding. arXiv preprint arXiv:2011.13534.

2) Zhelezniakov, D., Zaytsev, V., & Radyvonenko, O. (2021). Online handwritten mathematical expression recognition and applications: A survey. IEEe Access, 9, 38352-38373. **[1]**

3) Aggarwal, R., Pandey, S., Tiwari, A. K., & Harit, G. (2022). Survey of mathematical expression recognition for printed and handwritten documents. IETE Technical Review, 39(6), 1245-1253. **[9]**

4) Truong, T. N., Nguyen, C. T., Zanibbi, R., Mouchère, H., & Nakagawa, M. (2024). A survey on handwritten mathematical expression recognition: The rise of encoder-decoder and GNN models. Pattern Recognition, 110531.

5.2. STR paper

1) Fujitake, M. (2024). Dtrocr: Decoder-only transformer for optical character recognition. In Proceedings of the IEEE/CVF winter conference on applications of computer vision (pp. 8025-8035). **[2]**

5.3. PMER benchmark dataset

1) InftyMDB-1

https://www.inftyproject.org/en/index.html

2) Im2Latex 90K, 100K

Deng, Y., Kanervisto, A., Ling, J., & Rush, A. M. (2017, July). Image-to-markup generation with coarse-to-fine attention. In International Conference on Machine Learning (pp. 980-989). PMLR.

3) I2L-140K

Singh, S. S. (2018). Teaching machines to code: neural markup generation with visual attention. arXiv preprint arXiv:1802.05415.

4) FormulaNet

Schmitt-Koopmann, F. M., Huang, E. M., Hutter, H. P., Stadelmann, T., & Darvishy, A. (2022). FormulaNet: A benchmark dataset for mathematical formula detection. IEEE Access, 10, 91588-91596.

5) im2latexv2, realFormula

Schmitt-Koopmann, F. M., Huang, E. M., Hutter, H. P., Stadelmann, T., & Darvishy, A. (2024). MathNet: a data-centric approach for printed mathematical expression recognition. IEEE Access.

6) AMME-20K Dataset

Zong, T., Yang, L., Luo, W., Yu, W., Cai, B., Zhang, H., ... & Gao, L. (2024, November). ICPR 2024 Competition on Multi-line Mathematical Expressions Recognition. In International Conference on Pattern Recognition (pp. 17-29). Cham: Springer Nature Switzerland.

5.4. HMER benchmark dataset

1) Aida Calculus Math Handwriting Recognition Dataset

https://www.kaggle.com/datasets/aidapearson/ocr-data

2) MLHME-38K Dataset

https://ai.100tal.com/icdar

3) CROHME 2014, 2016, 2019, 2023 **[5]**

Mouchere, H., Viard-Gaudin, C., Zanibbi, R., & Garain, U. (2014, September). ICFHR 2014 competition on recognition of on-line handwritten mathematical expressions (CROHME 2014). In 2014 14th International Conference on Frontiers in Handwriting Recognition (pp. 791-796). IEEE.

Mouchère, H., Viard-Gaudin, C., Zanibbi, R., & Garain, U. (2016, October). ICFHR2016 CROHME: Competition on recognition of online handwritten mathematical expressions. In 2016 15th International Conference on Frontiers in Handwriting Recognition (ICFHR) (pp. 607-612). IEEE.

Mahdavi, M., Zanibbi, R., Mouchere, H., Viard-Gaudin, C., & Garain, U. (2019, September). ICDAR 2019 CROHME+ TFD: Competition on recognition of handwritten mathematical expressions and typeset formula detection. In 2019 International Conference on Document Analysis and Recognition (ICDAR) (pp. 1533-1538). IEEE.

Xie, Y., Mouchère, H., Simistira Liwicki, F., Rakesh, S., Saini, R., Nakagawa, M., ... & Truong, T. N. (2023, August). Icdar 2023 crohme: Competition on recognition of handwritten mathematical expressions. In International Conference on Document Analysis and Recognition (pp. 553-565). Cham: Springer Nature Switzerland.

4) HME 100K

Yuan, Y., Liu, X., Dikubab, W., Liu, H., Ji, Z., Wu, Z., & Bai, X. (2022). Syntax-aware network for handwritten mathematical expression recognition. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (pp. 4553-4562).

5) M^2E

Yang, W., Li, Z., Peng, D., Jin, L., He, M., & Yao, C. (2023, October). Read ten lines at one glance: line-aware semi-autoregressive Transformer for multi-line handwritten mathematical expression recognition. In Proceedings of the 31st ACM International Conference on Multimedia (pp. 2066-2077).

6) MathWriting

Gervais, P., Fadeeva, A., & Maksai, A. (2024). Mathwriting: A dataset for handwritten mathematical expression recognition. arXiv preprint arXiv:2404.10690.

5.5. HMER model

1) DenseWAP (DenseWAP, DenseWAP-TD, DenseWAP-MSA, DenseWAP+CTC) (2018)

Zhang, J., Du, J., & Dai, L. (2018, August). Multi-scale attention with dense encoder for handwritten mathematical expression recognition. In 2018 24th international conference on pattern recognition (ICPR) (pp. 2245-2250). IEEE.

2) WAP (2019)

Wang, J., Du, J., Zhang, J., & Wang, Z. R. (2019, September). Multi-modal attention network for handwritten mathematical expression recognition. In 2019 International Conference on Document Analysis and Recognition (ICDAR) (pp. 1181-1186). IEEE.

3) Stroke extraction (2020)

Chan, C. (2020). Stroke extraction for offline handwritten mathematical expression recognition. IEEE Access, 8, 61565-61575.

4) TD (2020)

Zhang, J., Du, J., Yang, Y., Song, Y. Z., Wei, S., & Dai, L. (2020, November). A tree-structured decoder for image-to-markup generation. In International Conference on Machine Learning (pp. 11076-11085). PMLR.

5) WS-WAP (2020)

Truong, T. N., Nguyen, C. T., Phan, K. M., & Nakagawa, M. (2020, September). Improvement of end-to-end offline handwritten mathematical expression recognition by weakly supervised learning. In 2020 17th International Conference on Frontiers in Handwriting Recognition (ICFHR) (pp. 181-186). IEEE.

6) BTTR (2021)
Zhao, W., Gao, L., Yan, Z., Peng, S., Du, L., & Zhang, Z. (2021). Handwritten mathematical expression recognition with bidirectionally trained transformer. In Document analysis and recognition–ICDAR 2021: 16th international conference, Lausanne, Switzerland, September 5–10, 2021, proceedings, part II 16 (pp. 570-584). Springer International Publishing.
7) WAP + Transformer design; WAP + MHA, WAP + MHA + Stacked decoder (2021)
Ding, H., Chen, K., & Huo, Q. (2021). An encoder-decoder approach to handwritten mathematical expression recognition with multi-head attention and stacked decoder. In Document analysis and recognition–ICDAR 2021: 16th international conference, Lausanne, Switzerland, September 5–10, 2021, proceedings, part II 16 (pp. 602-616). Springer International Publishing.
8) ABM (2022)
Bian, X., Qin, B., Xin, X., Li, J., Su, X., & Wang, Y. (2022, June). Handwritten mathematical expression recognition via attention aggregation based bi-directional mutual learning. In Proceedings of the AAAI conference on artificial intelligence (Vol. 36, No. 1, pp. 113-121).
9) CAN (CAN-ABM, CAN-DWAP) (2022)
Li, B., Yuan, Y., Liang, D., Liu, X., Ji, Z., Bai, J., ... & Bai, X. (2022, October). When counting meets HMER: counting-aware network for handwritten mathematical expression recognition. In European conference on computer vision (pp. 197-214). Cham: Springer Nature Switzerland.
10) CoMER (2022)
Zhao, W., & Gao, L. (2022, October). Comer: Modeling coverage for transformer-based handwritten mathematical expression recognition. In European conference on computer vision (pp. 392-408). Cham: Springer Nature Switzerland.
11) SAN (2022)
Yuan, Y., Liu, X., Dikubab, W., Liu, H., Ji, Z., Wu, Z., & Bai, X. (2022). Syntax-aware network for handwritten mathematical expression recognition. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (pp. 4553-4562).
12) SS-TD (2022)
Lin, Z., Li, J., Yang, F., Huang, S., Yang, X., Lin, J., & Yang, M. (2022, November). Spatial attention and syntax rule enhanced tree decoder for offline handwritten mathematical expression recognition. In International Conference on Frontiers in Handwriting Recognition (pp. 213-227). Cham: Springer International Publishing.
13) TDv2 (2022)
Wu, C., Du, J., Li, Y., Zhang, J., Yang, C., Ren, B., & Hu, Y. (2022, June). Tdv2: A novel tree-structured decoder for offline mathematical expression recognition. In Proceedings

of the AAAI Conference on Artificial Intelligence (Vol. 36, No. 3, pp. 2694-2702).
14) TSDNet (2022)
Zhong, S., Song, S., Li, G., & Chan, S. H. G. (2022, October). A tree-based structure-aware transformer decoder for image-to-markup generation. In Proceedings of the 30th ACM International Conference on Multimedia (pp. 5751-5760).
15) GCN (2023)
Zhang, X., Ying, H., Tao, Y., Xing, Y., & Feng, G. (2023, June). General category network: Handwritten mathematical expression recognition with coarse-grained recognition task. In ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) (pp. 1-5). IEEE.
16) LAST (Line Aware Semi-autoregressive Transformer) (2023)
Yang, W., Li, Z., Peng, D., Jin, L., He, M., & Yao, C. (2023, October). Read ten lines at one glance: line-aware semi-autoregressive Transformer for multi-line handwritten mathematical expression recognition. In Proceedings of the 31st ACM International Conference on Multimedia (pp. 2066-2077).
17) BPD; Branch Parallel Decoding (BPD, BPD-Coverage) (2024)
Li, Z., Yang, W., Qi, H., Jin, L., Huang, Y., & Ding, K. (2024). A tree-based model with branch parallel decoding for handwritten mathematical expression recognition. Pattern Recognition, 149, 110220.
18) NAMER (2024)
Liu, C., Pan, J., Hu, J., Yin, B., Yin, B., Chen, M., ... & Liu, Q. (2024, September). NAMER: Non-Autoregressive Modeling for Handwritten Mathematical Expression Recognition. In European Conference on Computer Vision (pp. 273-291). Cham: Springer Nature Switzerland.
19) ICAL (2024)
Zhu, J., Gao, L., & Zhao, W. (2024, August). ICAL: Implicit Character-Aided Learning for Enhanced Handwritten Mathematical Expression Recognition. In International Conference on Document Analysis and Recognition (pp. 21-37). Cham: Springer Nature Switzerland.
20) PosFormer (2024) **[4]**
Guan, T., Lin, C., Shen, W., & Yang, X. (2024, September). PosFormer: recognizing complex handwritten mathematical expression with position forest transformer. In European Conference on Computer Vision (pp. 130-147). Cham: Springer Nature Switzerland.
21) TAMER (2025) **[3]**
Zhu, J., Zhao, W., Li, Y., Hu, X., & Gao, L. (2025, April). Tamer: Tree-aware transformer for handwritten mathematical expression recognition. In Proceedings of the AAAI Conference on Artificial Intelligence (Vol. 39, No. 10, pp. 10950-10958).

5.6. PMER model

1) WYGIWYS (2017)

Deng, Y., Kanervisto, A., Ling, J., & Rush, A. M. (2017, July). Image-to-markup generation with coarse-to-fine attention. In International Conference on Machine Learning (pp. 980-989). PMLR.

2) I2L (I2L-NOPOOL, I2L-STRIPS) (2018)

Singh, S. S. (2018). Teaching machines to code: neural markup generation with visual attention. arXiv preprint arXiv:1802.05415.

3) ConvMath (2021)

Yan, Z., Zhang, X., Gao, L., Yuan, K., & Tang, Z. (2021, January). ConvMath: a convolutional sequence network for mathematical expression recognition. In 2020 25th International Conference on Pattern Recognition (ICPR) (pp. 4566-4572). IEEE.

4) MathNet (2024)

Schmitt-Koopmann, F. M., Huang, E. M., Hutter, H. P., Stadelmann, T., & Darvishy, A. (2024). MathNet: a data-centric approach for printed mathematical expression recognition. IEEE Access.

5.7. Models for both PMER and HMER

1) PAL (2019) **[6]**

Wu, J. W., Yin, F., Zhang, Y. M., Zhang, X. Y., & Liu, C. L. (2019). Image-to-markup generation via paired adversarial learning. In Machine learning and knowledge discovery in databases: European conference, ECML PKDD 2018, Dublin, Ireland, September 10–14, 2018, Proceedings, Part I 18 (pp. 18-34). Springer International Publishing.

2) DLA (2020) **[7]**

Le, A. D. (2020). Recognizing handwritten mathematical expressions via paired dual loss attention network and printed mathematical expressions. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (pp. 566-567).

3) PAL-V2 (2020) **[8]**

Wu, J. W., Yin, F., Zhang, Y. M., Zhang, X. Y., & Liu, C. L. (2020). Handwritten mathematical expression recognition via paired adversarial learning. International Journal of Computer Vision, 128, 2386-2401.

5.8. Papers related to experiments

5.8.1. Contrastive learning

1) CCLSL (2022)

Lin, Q., Huang, X., Bi, N., Suen, C. Y., & Tan, J. (2022). Cclsl: Combination of contrastive learning and supervised learning for handwritten mathematical expression recognition. In

Proceedings of the Asian Conference on Computer Vision (pp. 3724-3739).
2) PrimCLR (PCL; Primitive Contrastive Learning) (2022)
Guo, H. Y., Wang, C., Yin, F., Liu, H. Y., Wu, J. W., & Liu, C. L. (2022, August). Primitive contrastive learning for handwritten mathematical expression recognition. In 2022 26th International Conference on Pattern Recognition (ICPR) (pp. 847-854). IEEE.
3) MoCoMER (2023)
Pramanik, S., Mitra, S., & Das, N. (2024, December). MoCoMER: A Self Supervised Representation Learning for Handwritten Mathematical Expression Recognition. In Proceedings of the Fifteenth Indian Conference on Computer Vision Graphics and Image Processing (pp. 1-10).

5.8.2. Augmentation

1) PGS; Pattern Generation Strategies (2019)
Le, A. D., Indurkhya, B., & Nakagawa, M. (2019). Pattern generation strategies for improving recognition of handwritten mathematical expressions. Pattern Recognition Letters, 128, 255-262.
2) SADA; Scale Augmentation and Drop Attention (2020)
Li, Z., Jin, L., Lai, S., & Zhu, Y. (2020, September). Improving attention-based handwritten mathematical expression recognition with scale augmentation and drop attention. In 2020 17th International Conference on Frontiers in Handwriting Recognition (ICFHR) (pp. 175-180). IEEE.